

3 RAM ECC monitoring (RAMECC)

3.1 Introduction

The STM32H742, STM32H743/753 and STM32H750 devices feature a RAM ECC monitoring unit (RAMECC). It provides a mean for application software to verify ECC status and execute service routines when an error occurs.

3.2 RAMECC main features

SRAM data are protected by ECC. The ECC mechanism is based on the SECDED algorithm. It supports single- and double-error detection, as well as single-error correction:

- 7 ECC bits are added per 32-bit word.
- 8 ECC bits are added per 64-bit word for AXI-SRAM and ITCM-RAM.

RAM data word integrity is checked at each memory read access, or partial RAM word write operation. Two cycles are required to perform a partial RAM word write (read-modify-write).

The RAMECC monitoring unit includes the following features:

- RAM ECC monitoring per domain
- RAM failing address/data identification

3.3 RAMECC functional description

3.3.1 RAMECC block diagram

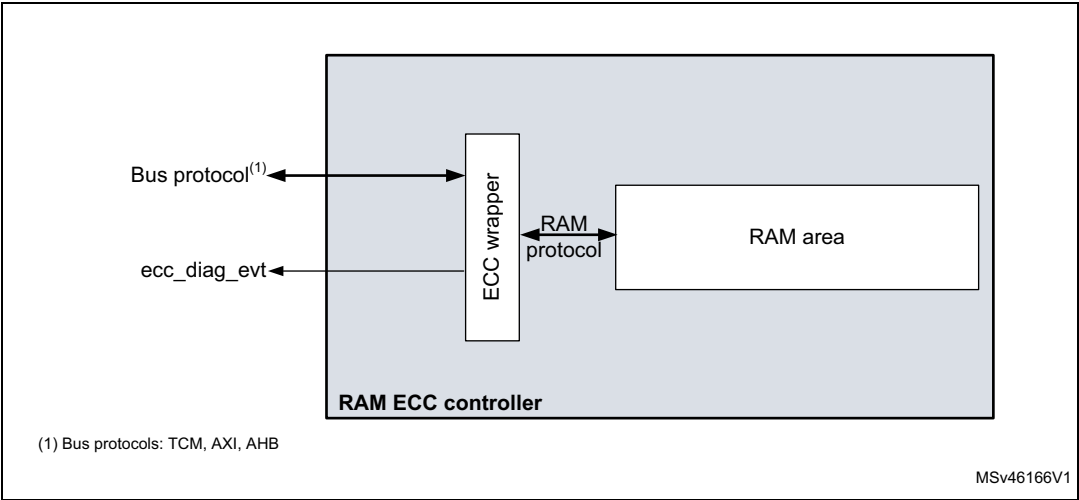
An ECC controller is associated to each RAM area. It performs the following functions:

- ECC encoding: ECC code computation and storage.
- ECC decoding: RAM data word loading and ECC code decoding to detect errors
- Error detection: single- and double-error detection
- Error correction: single-error correction.

Note: All the RAM ECC controllers are always enabled.

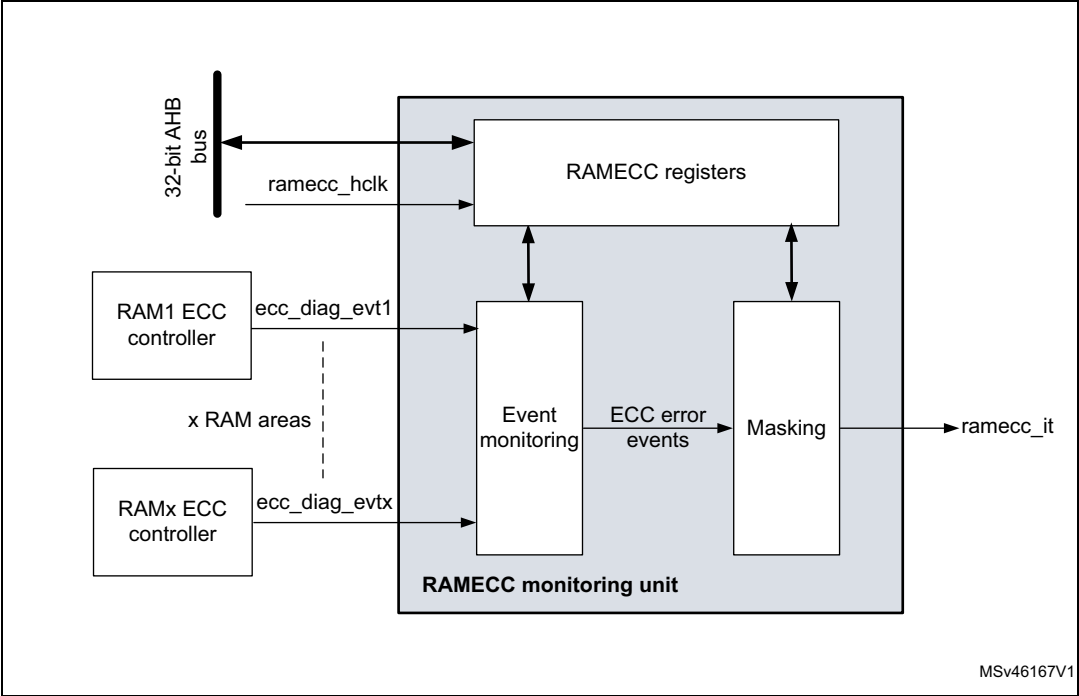
[Figure 3](#) describes the implementation of RAM ECC controllers.

Figure 3. RAM ECC controller implementation schematic



A dedicated RAM ECC monitoring area is defined for each domain (see [Section 3.3.3: RAMECC monitor mapping](#)). The RAMECC allows the collection of ECC diagnostic events from each RAM ECC controller and provides a mean for the CPU to verify the ECC status. [Figure 4](#) shows the connection schematic between the RAM ECC controller and the RAMECC monitoring unit.

Figure 4. Connection between RAM ECC controller and RAMECC monitoring unit



3.3.2 RAMECC internal signals

[Table 10](#) gives the list of the internal signals that control the RAMECC unit.

Table 10. RAMECC internal input/output signals

Internal signal name	Signal type	Description
ramecc_hclk	Input	AHB clock
ecc_diag_evtx	Input	ECC diagnostic event generated by RAMx ECC controller x
ramecc_it	Output	Interrupt generated by the RAMECC monitoring unit when an ECC error is detected.

3.3.3 RAMECC monitor mapping

STM32H742, STM32H743/753 and STM32H750 devices features three RAMECC monitoring units (one per domain). The inputs from the ECC controllers are mapped as described in [Table 11](#). The RAM ECC event monitoring status and configuration registers are described in [Section 3.4: RAMECC registers](#).

Table 11. ECC controller mapping

RAMECC units	Monitor number	SRAM ECC event monitoring status and configuration registers		Size in Kbytes	Address Offset
D1 domain RAMECC unit	1	AXI SRAM ECC monitoring unit		512	0x20
	2	ITCM-RAM ECC monitoring unit		64	0x40
	3	DTCM-RAM ECC monitoring unit	D0TCM	64	0x60
	4		D1TCM	64	0x80
	5	ETM RAM ECC monitoring unit		4	0xA0
D2 domain RAMECC unit	1	SRAM1 ECC monitoring unit	SRAM1_0	64	0x20
	2		SRAM1_1	64	0x40
	3	SRAM2 ECC monitoring unit	SRAM2_0	64	0x60
	4		SRAM2_1	64	0x80
	5	SRAM3 ECC monitoring unit		32	0xA0
D3 domain RAMECC unit	1	SRAM4 ECC monitoring unit		64	0x20
	2	Backup RAM ECC monitoring unit		4	0x40

3.4 RAMECC registers

RAMECC registers can be accessed only in 32-bit (word) mode. Byte and half-word formats are not allowed.

3.4.1 RAMECC interrupt enable register (RAMECC_IER)

Address offset: 0x00

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	GECCDEBWIE	GECCDEIE	GECCSEIE	GIE
												rw	rw	rw	rw

Bits 31:4 Reserved, must be kept at reset value.

Bit 3 **GECCDEBWIE**: Global ECC double error on byte write (BW) interrupt enable

When GECCDEBWIE bit is set to 1, an interrupt is generated when an ECC double detection error occurs during a byte write operation to RAM (incomplete word write).

0: no interrupt generated when an ECC double detection error occurs on byte write
1: interrupt generated if an ECC double detection error occurs on byte write

Bit 2 **GECCDEIE**: Global ECC double error interrupt enable

When GECCDEIE bit is set to 1, an interrupt is generated when an ECC double detection error occurs during a read operation from RAM.

0: no interrupt generated when an ECC double detection error occurs
1: interrupt generated if an ECC double detection error occurs

Bit 1 **GECCSEIE**: Global ECC single error interrupt enable

When GECCSEIE bit is set to 1, an interrupt is generated when an ECC single error occurs during a read operation from RAM.

0: no interrupt generated when an ECC single error occurs
1: interrupt generated when an ECC single error occurs

Bit 0 **GIE**: Global interrupt enable

When GIE bit is set to 1, an interrupt is generated when an enabled global ECC error (GECCDEBWIE, GECCDEIE or GECCSEIE) occurs.

0: no interrupt generated when an ECC error occurs
1: interrupt generated when an ECC error occurs

3.4.2 RAMECC monitor x configuration register (RAMECC_MxCR)

Address offset: $0x20 * x$

Reset value: 0x0000 0000

x is the ECC monitoring unit number

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ECCEL EN	ECCDE BWIE	ECCDE IE	ECCSE IE	Res.	Res.
										rw	rw	rw	rw		

Bits 31:6 Reserved, must be kept at reset value.

Bit 5 **ECCELEN**: ECC error latching enable

When ECCELEN bit is set to 1, if an ECC error occurs (both for single error correction or double detection) during a read operation, the context (address, data and ECC code) that generated the error are latched to their respective registers.

0: no error context preserved when an ECC error occurs

1: error context preserved when an ECC error occurs

Bit 4 **ECCDEBWIE**: ECC double error on byte write (BW) interrupt enable

When ECCDEBWIE bit is set to 1, monitor x generates an interrupt when an ECC double detection error occurs during a byte write operation to RAM.

0: no interrupt generated when an ECC double detection error occurs on byte write

1: interrupt generated if an ECC double detection error occurs on byte write

Bit 3 **ECCDEIE**: ECC double error interrupt enable

When ECCDEIE bit is set to 1, monitor x generates an interrupt when an ECC double detection error occurs during a read operation from RAM.

0: no interrupt generated when an ECC double detection error occurs

1: interrupt generated if an ECC double detection error occurs

Bit 2 **ECCSEIE**: ECC single error interrupt enable

When ECCSEIE bit is set to 1, monitor x generates an interrupt when an ECC single error occurs during a read operation from RAM.

0: no interrupt generated when an ECC single error occurs

1: interrupt generated when an ECC single error occurs

Bits 1:0 Reserved, must be kept at reset value.

3.4.3 RAMECC monitor x status register (RAMECC_MxSR)

Address offset: $0x24 + 0x20 * (x - 1)$, (x= ECC monitoring unit number)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	DEBW DF	DEDF	SEDCF
													rc_w0	rc_w0	rc_w0

Bits 31: 3 Reserved, must be kept at reset value.

Bit 2 **DEBWDF**: ECC double error on byte write (BW) detected flag

This bit is set by hardware. It is cleared by software by writing a 0

0: no error detected

1: error detected

Bit 1 **DEDF**: ECC double error detected flag

This bit is set by hardware. It is cleared by software by writing a 0

0: no error detected

1: error detected

Bit 0 **SEDCF**: ECC single error detected and corrected flag

This bit is set by hardware. It is cleared by software by writing a 0

0: no error detected and corrected

1: error detected and corrected

3.4.4 RAMECC monitor x failing address register (RAMECC_MxFAR)

Address offset: $0x28 + 0x20 * (x-1)$, (x = ECC monitoring unit number)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FADD[31:16]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FADD[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:0 **FADD[31:0]**: ECC error failing address

When an ECC error occurs the FADD bitfield contains the address that generated the ECC error.

3.4.5 RAMECC monitor x failing data low register (RAMECC_MxFDRL)

Address offset: $0x2C + 0x20 * (x-1)$, (x = ECC monitoring unit number)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FDATA[31:16]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FDATA[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:0 **FDATA[31:0]**: Failing data low

When an ECC error occurs the FDATA bitfield contains the LSB part of the data that generated the error. For 32-bit word SRAM, this bitfield contains the full memory word that generated the error.

3.4.6 RAMECC monitor x failing data high register (RAMECC_MxFDRH)

Address offset: $0x30 + 0x20 * (x-1)$, (x= ECC monitoring unit number)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FDATAH[31:16]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FDATAH[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:0 **FDATAH[31:0]**: Failing data high (64-bit memory)

When an ECC error occurs the FDATAH bitfield contains the MSB part of the data that generated the error.

Note: This register is reserved in case of 32-bit word SRAM.

3.4.7 RAMECC monitor x failing ECC error code register (RAMECC_MxFECR)

Address offset: $0x34 + 0x20 * (x-1)$, (x= ECC monitoring unit number)

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
FEC[31:16]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FEC[15:0]															
r	r	r	r	r	r	r	r	r	r	r	r	r	r	r	r

Bits 31:0 **FEC [31:0]**: Failing error code

When an ECC error occurs the FEC bitfield contains the ECC failing code that generated the error.

3.4.8 RAMECC register map

Table 12. RAMECC register map and reset values

Offset	Register name reset value	Register size																																	
		31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0		
0x00	RAMECC_IER	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	GECCDEB	WIE	0	0	0
	Reset value																														0	0	0	0	0
0x20 * x (x = monitoring unit number)	RAMECC_MxCR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res
	Reset value																																		
0x24+0x20 * (x -1) (x = monitoring unit number)	RAMECC_MxSR	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	Res	
	Reset value																																		
0x28+0x20 * (x -1) (x = monitoring unit number)	RAMECC_MxFAR	FADD[31:0]																																	
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x2C+0x20 * (x -1) (x = monitoring unit number)	RAMECC_MxFDRL	FDATA[31:0]																																	
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x30+0x20 * (x -1) (x = monitoring unit number)	RAMECC_MxFDRH	FDATAH[31:0]																																	
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0x34+0x20 * (x -1) (x = monitoring unit number)	RAMECC_MxFECR	FEC[31:0]																																	
	Reset value	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Refer to [Section 2.3 on page 128](#) for the register boundary addresses.